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Interpreting and Improving Large Language Models in Arithmetic Calculation

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Abstract

Large language models (LLMs) have demonstrated remarkable potential across numerous applications and have shown an emergent ability to tackle complex reasoning tasks, such as mathematical computations. However, even for the simplest arithmetic calculations, the intrinsic mechanisms behind LLMs remain mysterious, making it challenging to ensure reliability. In this work, we delve into uncovering a specific mechanism by which LLMs execute calculations. Through comprehensive experiments, we find that LLMs frequently involve a small fraction ($< 5\%$) of attention heads, which play a pivotal role in focusing on operands and operators during calculation processes. Subsequently, the information from these operands is processed through multi-layer perceptrons (MLPs), progressively leading to the final solution. These pivotal heads/MLPs, though identified on a specific dataset, exhibit transferability across different datasets and even distinct tasks. This insight prompted us to investigate the potential benefits of selectively fine-tuning these essential heads/MLPs to boost the LLMs' computational performance. We empirically find that such precise tuning can yield notable enhancements on mathematical prowess, without compromising the performance on non-mathematical tasks. Our work serves as a preliminary exploration into the arithmetic calculation abilities inherent in LLMs, laying a solid foundation to reveal more intricate mathematical tasks.

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1. Introduction

Large language models (LLMs) have experienced rapid advancements and shown impressive language understanding capabilities (Devlin et al., 2019; Brown et al., 2020; Chowdhery et al., 2022). Notably, LLMs exhibit emergent abilities (Wei et al., 2022b) that enable them to solve intricate reasoning tasks akin to humans, such as mathematical computations (Frieder et al., 2023; Jie et al., 2022), chain-of-thought reasoning (Wei et al., 2022c; Kojima et al., 2022), few-shot prompting (Brown et al., 2020; Alayrac et al., 2022), etc. Despite these impressive characteristics, the complex inner processes governing LLMs' functionality have yet to be fully illuminated, due to the complex and intricate non-linear interactions within densely-connected layers. Comprehending these underlying mechanisms could contribute to predicting how the LLMs behave beyond their training data (Mu & Andreas, 2020), gaining insights into the emergence of certain behaviors (Nanda & Lieberum, 2022; Barak et al., 2022; Wei et al., 2022a), as well as identifying and rectifying errors present in the specific models (Hernandez et al., 2021; Vig et al., 2020).

In this work, we take the first attempt to interpret the inner process of LLMs through the lens of mathematical computation problems, which are conducted on publicly available LLMs (*e.g.*, LLaMA2 series (Touvron et al., 2023b)). Unlike typical language comprehension tasks, mathematical computation tasks involve concise problem statements with definitive correct answers, requiring a process of reasoning and calculation rather than direct copying to derive the solutions. These characteristics enable us to gain insights into the models' reasoning capabilities without interference from unrelated factors. Specifically, we focus on tasks involving the arithmetic calculation with two operands, *i.e.*, addition, subtraction, multiplication, and division, which are fundamentals of mathematical computation. To this end, we create datasets of various types of sentences that involve the calculation logic, such as "The addition of 3 and 5 equals to _" in Figure 1. The LLMs could provide answers with high confidence scores of over 80% on average.

To unveil how these models correctly complete the task (*e.g.*, "$3 + 5 = 8$"), we begin by identifying the task-related internal components in LLMs. We do a hard in-

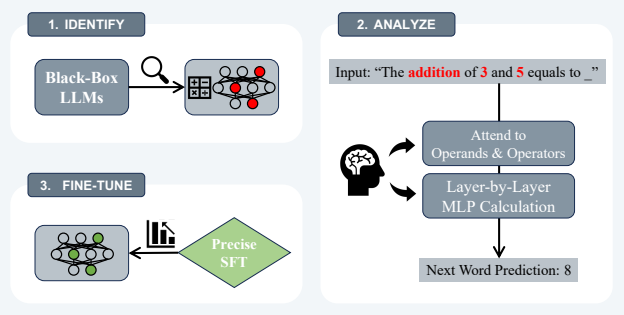


Figure 1: The pipeline involves three steps: 1) *identify* the key components attributed to arithmetic calculations in black-box LLMs, 2) *analyze* the working mechanism of the key components towards human-understandable explanations, 3) *fine-tune* the key components to precisely improve the mathematical capability of LLMs.

tervention (Pearl, 2009) on the transformer attention heads and multi-layer perceptrons (MLPs) to observe their effects on the predicted logits[1]. Our findings reveal that only a small percentage ($< 5\%$) of the attention heads and the MLPs after these heads significantly impact the model's performance. Namely, LLMs frequently involve these attention heads and the subsequent MLPs when completing the calculations. Subsequently, we knock out these frequently-involved heads/MLPs to validate their faithfulness. We find that the model performance decreases sharply when those pivotal heads/MLPs are knocked out, resulting in a decrease of around 70% in accuracy.

To interpret the working mechanism of identified heads/MLPs towards human-understandable explanations, we gain a deeper analysis of their operational "behaviors". Specifically, we investigate the attention patterns of the crucial heads, and find that these attention heads exhibit a strong focus on the tokens representing operands and operators within mathematical sentences, demonstrating a relative insensitivity to other non-relevant tokens. For the analysis of MLPs, we compare the correlations between the embeddings of MLPs' input/output and the embeddings of number tokens (*i.e.*, operands and answers). It reveals that the MLPs, guided by these number-attended heads, take operands as input, and mirror the attributes of tokens corresponding to correct answers more closely. These observations lead us to hypothesize that *LLMs may initially employ a set of heads to pinpoint arithmetic operands from text, subsequently engaging MLPs to work out the answers.* Additionally, the observed behaviors of these heads/MLPs exhibit a high degree of transferability, analogous to adversarial examples being *transferable across models* (Szegedy et al., 2014). Namely, the key heads/MLPs identified on one dataset are also effective for other datasets. For instance, their impact is noticeable on the publicly available math datasets (*e.g.*, SVAMP (Patel et al., 2021)), as well as varied data formats involving multi-digit integers, rational numbers, etc. This empirical observation underscores the crucial role of key heads/MLPs in mathematical calculations.

In addition to uncovering the internal mechanisms, we have devised an effective strategy that involves targeted fine-tuning of the specific attention heads and MLPs closely tied to mathematical computations, thereby enhancing the model's mathematical prowess. The experimental results are compelling: with fine-tuning as few as 32 attention heads (with a total of 1024 heads), we observe a remarkable improvement in the model's mathematical capabilities. This precise tuning methodology not only matches but can surpass the enhancements achieved through full-model fine-tuning. Moreover, this fine-grained strategy of adjustment has a distinct advantage—it leaves most of the model's parameters unchanged, avoiding the performance trade-offs in non-mathematical domains commonly observed with full-model fine-tuning.

In summary, this work aims to delve into the inner mechanism of LLMs through mathematical calculation tasks, along the pipeline of "identify-analyze-finetune" shown in Figure 1. Our findings reveal a sparsity in the attention heads of LLMs, with less than 5% of heads exhibiting close correlations. These heads particularly attend to the operands and operators, while the subsequent MLPs gradually deduce the correct answers. The discovered mechanism shows strong cross-dataset transferability and inspires us to precisely finetune the calculation-related heads/MLPs for better mathematical capability. We empirically find that precise tuning brings in much less impact on non-mathematical tasks when improving the targeted ability of LLMs.

2. Related Works

Interpretability Methods. Interpreting the inner mechanism of large language models (LLMs) has become increasingly urgent in recent years (Madsen et al., 2022; Rauker et al., 2023), especially when LLMs are applied in high-stakes decision-making domains such as healthcare, criminal justice, and finance (Obermeyer et al., 2019; Rudin, 2019; Bender et al., 2021). Vig et al. (2020) adapted the approach of *causal mediation analysis* (CMA) (Pearl, 2001) for interpreting the deep language models, and it has been applied for various tasks, such as subject-verb agreement (Finlayson et al., 2021), natural language inference (Geiger et al., 2021), retention of factual associations (Meng et al., 2022; Geva et al., 2023). Furthermore, *path patching* extends the concept of CMA by measuring how a treatment

[1]Here, doing a hard intervention is equal to replacing the value of attention heads and MLPs, while performing a soft intervention means modifying the modules for calculating the attention and MLP values (Pearl, 2009).

effect is mediated by node-to-node connections between individual neurons or features. Recent works have used path patching to explain neural networks in terms of circuits (Olah et al., 2023), identified for different capabilities including indirect object identification (Wang et al., 2023a), greater-than computation (Hanna et al., 2023), and mapping answer text to answer labels (Lieberum et al., 2023).

Interpretability for Mathematical Tasks. Mathematical ability has long been a subject of interest in natural language processing (Kushman et al., 2014; Huang et al., 2016; Wang et al., 2017; Thawani et al., 2021). Some studies have investigated the mathematical abilities of LLMs (Frieder et al., 2023; Saxton et al., 2019; Nogueira et al., 2021; Qian et al., 2023; Imani et al., 2023; Romera-Paredes et al., 2024), but they mainly focus on explaining *what* these models can do rather than *how* they do it. In contrast, some other studies have dived deeper into the LLM structure without treating LLM as an inscrutable black box. Stolfo et al. (2023) identified the key attention *layers* relating to arithmetic questions, but lacking in-depth explanation and validation of the key layers' behaviors. Wu et al. (2023) scaled the methods from causal abstraction to understand how Alpaca (7B) (Taori et al., 2023) follows the instruction in comparing two numbers. (Hanna et al., 2023) provided a causal explanation about how GPT2-small (0.1B) (Radford et al., 2019) implements the "greater-than" task, but only reveal simple phenomena limited by the small size of model and the lack of diversity in the dataset.

Fine-tune LLMs for Mathematical Tasks. Numerous studies improve the mathematical reasoning ability of LLMs by aggregating various sampled reasoning paths during either fine-tuning or inference. Cobbe et al. (2021) train and devise a reasoning path verifier to select the correct results during inference. Wang et al. (2023b) propose to sample various reasoning paths during inference and then derive the final result by majority voting on the answers or through verifiers (Li et al., 2023). Uesato et al. (2022) explore to use of reinforcement learning methods for improving the mathematical reasoning abilities of LLMs. Several works apply the idea of rejection sampling along with other techniques to filter the diverse sampled reasoning paths for fine-tuning data augmentation (Huang et al., 2022; Zelikman et al., 2022; Ni et al., 2023). There also exist related works (Panigrahi et al., 2023) that locate key parameters to update for better task-specific ability. Panigrahi et al. (2023) locates a minuscule subset of parameters from an already fine-tuned model onto a pre-trained model without further tuning. The selection process for this subset is via optimizing the task-related objective function with L1 norm ensuring the sparsity of the subset. In our work, we locate the task-related parameters of pre-trained model via measuring the *causal effect* of each component, then *precisely fine-tune* the key components for mathematical tasks.

3. Preliminary

Large Language Models (LLMs). The LLMs utilized in this work comprise LLaMA2-7B and LLaMA2-13B (Touvron et al., 2023a). These are pre-trained language models freely available from HuggingFace[2]. All of these models are decoder-only transformers equipped with multi-head attention (MHA) and a single MLP in one transformer layer. For example, LLaMA2-7B consists of 32 transformer layers and 32 attention heads in MHA for each layer.

Transformer Architecture. The input to the transformer is a combination of position and token embeddings in $\mathbb{R}^{N\times d}$, where N is the number of tokens in the input and d is the model dimension. Following the definitions in (Elhage et al., 2021), the input embedding serves as the initial value for the *residual stream*, which is read from and written to by all attention heads and MLPs. Focusing on individual heads, the j-th head in the i-th layer is parametrized by four matrices: $W_Q^{i,j}$, $W_K^{i,j}$, $W_V^{i,j} \in \mathbb{R}^{d\times\frac{d}{H}}$, and $W_O^{i,j} \in \mathbb{R}^{\frac{d}{H}\times d}$. To simplify these parameters, we can express them as low-rank matrices in $\mathbb{R}^{d\times d}$: $W_{OV}^{i,j} = W_O^{i,j}W_V^{i,j}$ and $W_{QK}^{i,j} = W_Q^{i,j}(W_K^{i,j})^T$. The QK matrix is used to compute the attention pattern $A_{i,j} \in \mathbb{R}^{N\times N}$ for head (i, j), while the OV matrix determines the information written into the residual stream. At the end of the forward pass, a layer norm is applied before the unembed matrix W_U projects the residual stream into logits.

Task and Dataset. We focus on classic and widely encountered mathematical operations, *e.g.*, addition, subtraction, multiplication, division. Taking addition as an example, the arithmetic logic of addition ({A} + {B} = {C}) might naturally appear in sentences. Taking inspiration from the sentence styles and forms present in mathematical benchmarks of GSM8K (Cobbe et al., 2021) and SVAMP (Patel et al., 2021), we create a dataset for the addition task containing $10,000$ samples based on 36 templates with random single-token names, objects, and numbers. To assess the performance of LLMs on the calculation task, we measure the prediction probability of the {C} token. The average probability of correct predictions across the models was 82%. In this study, we select the samples that the language models are able to predict correctly. We denote the sentences generated by this procedure as reference data using the notation of X_{r}. For the templates and sentences, please refer to Figure 8 and Figure 10 in Appendix A.

Moreover, to meet the demand for perturbing component activation, we create another dataset comprising counterfactual sentences without the inclusion of calculation logic, using the notation of X_{c}. The samples are generated following two core principles: (1) maintaining the grammatical structures derived from the X_{r} templates; (2) substituting

[2]https://huggingface.co/